DINUKA EKANAYAKE

Software Engineer

Full-stack software engineer with hands-on experience building scalable web applications, AR-based interfaces, and applied AI systems. Delivered production-grade features in ERP systems during industry internships and built an AI-powered virtual try-on platform integrating AR, segmentation, and diffusion models. Passionate about crafting user-focused digital experiences, solving complex engineering problems, and continuously expanding expertise across full-stack development and machine learning.

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WORK EXPERIENCE

Software Engineer

Csquare Technologies Private Limited | Jun 2023 - Jun 2025

Worked as a Software Engineer at Csquare Technologies Private Limited since June 2023. As an Intern - Software Engineer and Software Engineer, I had responsibilities to develop, maintain, and test ongoing web-based ERP systems developed by the company.

- **ERP Developments**: Designing, coding, and testing robust features and modules for companies' internal and external ERP systems.
- **System Analysis and Design**: Excelling in requirements gathering, translating complex business needs into clear, technical specifications, and designing scalable system architectures.
- **System Implementation**: Overseeing the successful deployment of new ERP solutions and updates, ensuring smooth transition and minimal disruption to operations.
- **Customer Support**: Providing expert, high-level technical support, resolving critical post-implementation issues, and ensuring high customer satisfaction.

Mainly working with : PHP, HTML, CSS, JS

SKILLS

Programming

Python, TypeScript, JavaScript, PHP, Java, C/ C++

Frontend Development

HTML5, Three.js, CSS, Tailwind CSS, Next.js, React

Backend Development & APIs

Node.js, Laravel, Flask, FastAPI, REST API Design & Integration

Databases

MySQL, MongoDB, Firebase

Cloud, DevOps & Deployment

AWS, Azure, Docker, Kubernetes, Jenkins, Terraform, Linux, Git, GitHub, CI/CD

AI / Machine Learning

TensorFlow/Keras, U^2-Net, MediaPipe Pose, OpenCV, Image Classification, Segmentation, Model Deployment

Tools & Development Workflow

VS Code, Figma, Adobe Photoshop, Android Studio, Agile/Scrum, Version Control

EDUCATION

BSc (Hons) in Engineering in Information and Communication Engineering (Software Engineering)

Sri Lanka Technology Campus | 2019 - 2025

- Specializing Software Dev & Design | OOP | DS & Algorithms | DBMS | Cloud | Data Science | Telecommunication | ML DL NN

Advanced Level in Physical Science Stream

Ranabima Royal College | 2008 - 2016

PROJECTS

Hybrid AR + AI Virtual Try-On System for Fashion E-Commerce

Built a dual-workflow virtual try-on platform that allows users to preview garments in real-time AR or generate high-fidelity AI try-on results from photos.
Designed a complete pipeline including garment classification (CNN), background removal (U^2-Net), AR overlay using MediaPipe Pose, and a diffusion-based virtual try-on engine (CatVTON) deployed on Hugging Face.

Tech: Next.js, FastAPI, Python, TensorFlow/Keras, U^2-Net, MediaPipe, Cloudinary, Gradio, Hugging Face Spaces

Developer Portfolio Website (Next.js + Modern UI Engineering)

Designed and developed a modern portfolio website with interactive animations and performance-focused UI architecture.

- Built with React, TypeScript, and Vite using reusable components
- Created smooth UI interactions using Framer Motion and Canvas-based particle effects
- Implemented responsive design with Tailwind CSS and centralized content architecture
- Configured production deployment on GitHub Pages.

Tech: React, TypeScript, Vite, Tailwind CSS, Framer Motion, Canvas API

Multi-Tenant CRM System (React + Full-Stack Architecture)

Built a full-stack multi-tenant CRM platform with secure organization-level data isolation and role-based access control.

- Developed a React (Vite) dashboard with responsive UI and role-based feature visibility
- Implemented JWT authentication, RBAC, and activity logging for secure workflows
- Integrated REST APIs with Django REST Framework and PostgreSQL
- Used AWS S3 for secure file uploads and asset storage

Tech: React, Vite, TypeScript, Axios, React Router, Django REST Framework, PostgreSQL, AWS S3, JWT

Interactive 3D Virtual Art Gallery (Web-Based)

I created a fully interactive 3D virtual art gallery where users can walk through a gallery space, view paintings in a realistic environment, and explore digitally placed sculptures. The system features custom room layouts, dynamic artwork placement, real-time navigation, textured walls/floors, and 3D statue models rendered using Three.js.

Tech: JavaScript, Three.js, WebGL, HTML/CSS

Moodify – Emotion-Based Music Recommendation Web App

I developed a web application that detects the user's emotional state and recommends personalized music playlists using real-time emotion analysis and the Spotify API. The app integrates a machine learning model with a Flask backend and features a responsive frontend for smooth user interaction.

Tech: Python, Flask, TensorFlow/Keras, JavaScript, HTML/CSS, Spotify API

Agri Mart Android Application Project

Agri Mart is an online shopping app that creates a digital market for all kinds of agricultural products. Everyone can sell their products and buy products through this application.

Tech: Java, Firebase

Advanced ToDo Application

- Built a feature-rich task management application in React with efficient state handling and responsive UI updates
- Implemented full task lifecycle (create, edit, delete, complete) with priority levels, due dates, and dynamic search/filtering
- Designed a modular component-based architecture using React Hooks for predictable and maintainable state management
- Integrated Local Storage for data persistence and developed a responsive UI with dark mode support using Tailwind CSS

Tech: React, JavaScript, Tailwind CSS, Local Storage API

STRENGTHS

- Creative Problem-Solving
- Critical Thinking
- Leadership
- Flexibility
- Communication
- Team Collaboration
- Time Management
- Adaptability

CERTIFICATIONS

AWS Academy Graduate - AWS Academy Cloud Foundations
Amazon Web Services (AWS)

AWS Academy Graduate - AWS Academy Cloud Architecting
Amazon Web Services (AWS)

KodeKloud Engineer - Docker (Level 1)
KodeKloud

KodeKloud Engineer - Linux (Level 1)
KodeKloud

KodeKloud Engineer - Jenkins (Level 1)
KodeKloud

KodeKloud Engineer - Kubernetes (Level 1)
KodeKloud

KodeKloud Engineer - Ansible (Level 1)
KodeKloud

KodeKloud Engineer - Git (Level 1)
KodeKloud

VOLUNTEER EXPERIENCE

Vice Chairman
IEEE Computer Society Student Branch Chapter of SLTC | 2021 - 2022

Best Student Branch Chapter Recognition at IEEE Boost 2021 by IEEE Sri Lanka Section

Student Advisor - HackTron v1 .0 Project consisting
Phase 1 - 1O Day Game Development Bootcamp
Phase 2 - Inter-Uni Game Dev Hackathon (First Ever Hackathon by SLTC)
Phase 3 - RISE UP Industrial Talks

Student Advisor - Month of CodegeNiX 2021 Project, which was a program series consisting of
1) Panel Discussion on How to Become Successful in the IT Industry
2) Road to CODEMANIA 21' - Python Fundamentals and Algorithmic Workshop Series
3) AdaGrad 21' - Introduction to Machine Learning Workshop
4) CODEMANIA v2.0 - SLTC's First-Ever Algorithmic Programming Competition

Treasurer
Media Unit of SLTC | 2021 - 2022

Have completed several fundraising projects to collect funds for the Media Unit and held the position of Treasurer on several inter-university projects. Currently holding the position of Advisor to the Executive Committee.

Vice Chairman
CYBERHAT 1.0 project | Dec 2020 - jan 2021
IEEE Computer Society Student Branch Chapter of SLTC

"CYBERHAT 1.0 - Getting Started with Cybersecurity and Ethical Hacking" Project. The online workshop for those who are interested in cybersecurity & the ones who are searching for a cybersecurity career path.

Media and Publicity Team Deputy Head
Leo IEEE Sinhalen Docs Project | 2020 - 2021

An all-island youth empowerment project collaboratively done by the IEEE Student Branch of SLTC and the Leo Club of SLTC.

Photographer and Graphic Designer
Media Unit of SLTC | 2019 - 2022

Have covered more than 120 events as a photographer and worked as a graphic designer in media unit of SLTC.

REFERENCES

Mr. Chamara Rathnayake
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